Prabhavit Gupta

Forward Deployed Engineer | AI Search Integrations, Agent Tool Loops & Production Web Systems

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (EU Permanent Resident | Relocation Ready to SF | Immediate Availability)

Executive Summary

- **Forward Deployed Engineering & High Agency Delivery:** Full-stack AI engineer with 3 years commercial software experience taking user-facing applications from early architecture through production integration. High ownership engineer comfortable embedding directly with technical founders, CTOs, and developer teams.
- **Search, Agent Harnesses & API Integrations:** Hands-on background architecting agentic execution harnesses, web search retrieval layers, semantic filtering, and Model Context Protocol (MCP) tool servers. Experienced turning complex API endpoints into dependable production workflows with robust error recovery.
- **Full-Stack Prototyping & Commercial Impact:** Skilled shipping customer-facing demos, developer reference architectures, and production web services (TypeScript, Python, Next.js, Node.js). Experienced bridging customer feedback directly into platform enhancements.

Technical Skills & Core Competencies

- **AI Search & Agent Tooling:** Search API Integrations, Agent Tool Loops, Model Context Protocol (MCP), Semantic Retrieval, Web Scraping (Playwright, BeautifulSoup), Prompt Engineering, Structured Outputs (Pydantic, Zod).
- **Full-Stack Software Engineering:** TypeScript, Python, Node.js, Next.js, React, Laravel, REST APIs, GraphQL, Microservices, Async Processing, SQLite, PostgreSQL (pgvector).
- **Developer Platforms & Deployment:** Docker, GitHub Actions, CI/CD, Sandboxed Tool Execution, Webhook Architecture, Postman, Distributed Tracing (Langfuse).
- **Customer-Facing Engineering:** Technical Scoping, Rapid Prototyping, Live Integration Troubleshooting, Architecture Workshops, Technical Demos & Developer Documentation.

Professional Experience

Forward Deployed AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Partnered directly with engineering teams and technical leadership to design, prototype, and ship custom AI agent integrations and real-time knowledge retrieval pipelines.
- Built custom Model Context Protocol (MCP) tools and API connectors enabling autonomous agents to execute external searches, scrape live web content, and synthesize structured data.
- Developed full-stack demo applications and developer reference guides demonstrating production integration patterns, caching strategies, and rate-limit handling.
- Implemented automated evaluation suites and trace observability with Langfuse, isolating API degradation, latency bottlenecks, and retrieval inaccuracies.
- Conducted technical discovery and architecture reviews with executive stakeholders, unblocking mission-critical customer deployments.

Lead Developer & Product Engineer • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Scaled platform from 2,000 to 4,500+ active users, architecting core backend features, third-party integrations, and automated customer-facing agents.
- Led the migration of the live platform off a legacy stack onto Next.js and Laravel without downtime or service disruption for members.
- Deployed autonomous customer support agents querying live platform state and external booking APIs with strict validation boundaries and human-in-the-loop escalation.
- Built automated data ingestion and extraction pipelines aggregating unstructured venue data across multiple European markets.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Shipped automation services and API integrations in Python and TypeScript, connecting disparate enterprise systems and databases.
- Implemented continuous integration pipelines with automated test suites in Docker, reducing deployment cycle times.
- Collaborated directly with business end-users to understand operational workflows and translate requirements into reliable software tools.

Key Engineering Projects

- **Agentic Search & Knowledge Gateway (MCP + Python + TypeScript):** Implemented an autonomous agent tool harness connecting frontier LLMs to real-time search APIs, semantic indexing, and sandboxed browser scraping with strict execution safety.
- **AI Data Extraction & Ingestion Pipeline (Python + Playwright + Claude API):** Engineered an automated web crawling and structured extraction engine processing 900+ multi-source data records with automated validation.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous foundation in formal logic, epistemology, decision theory, and system analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Ready for international visa sponsorship (H-1B, O-1, etc.) and immediate relocation to San Francisco.